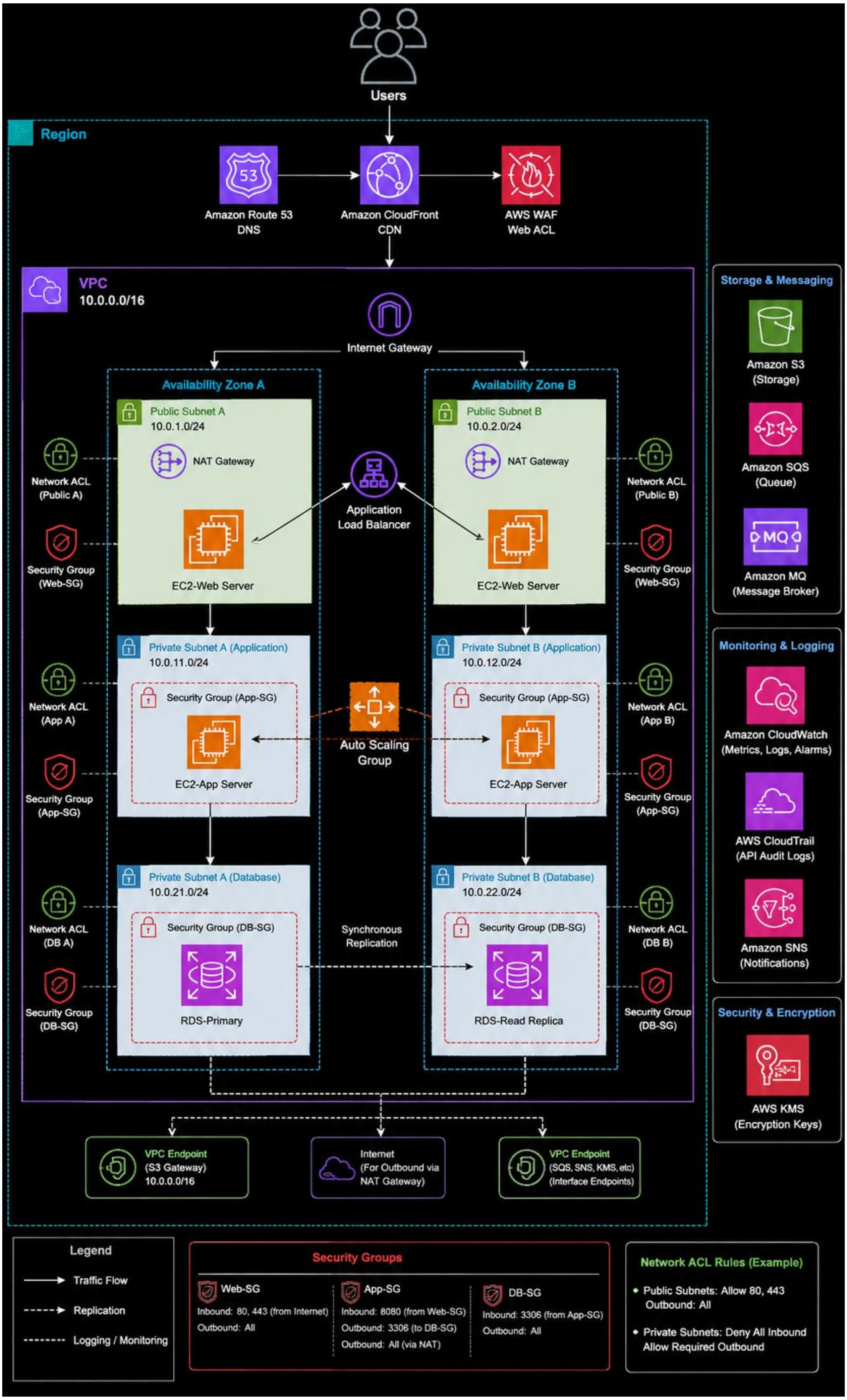


Users
Region
Amazon Route 53
DNS
Amazon CloudFront
CDN
AWS WAF
Web ACL
VPC
10.0.0.0/16
Internet Gateway
Availability Zone A
Availability Zone B
Public Subnet A
10.0.1.0/24
NAT Gateway
EC2-Web Server
Public Subnet B
10.0.2.0/24
NAT Gateway
EC2-Web Server
Application
Load Balancer
Network ACL
(Public A)
Security Group
(Web-SG)
Network ACL
(Public B)
Security Group
(Web-SG)
Private Subnet A (Application)
10.0.11.0/24
Security Group (App-SG)
EC2-App Server
Private Subnet B (Application)
10.0.12.0/24
Security Group (App-SG)
EC2-App Server
Auto Scaling
Group
Network ACL
(App A)
Security Group
(App-SG)
Network ACL
(App B)
Security Group
(App-SG)
Private Subnet A (Database)
10.0.21.0/24
Security Group (DB-SG)
RDS-Primary
Private Subnet B (Database)
10.0.22.0/24
Security Group (DB-SG)
RDS-Read Replica
Synchronous
Replication
Network ACL
(DB A)
Security Group
(DB-SG)
Network ACL
(DB B)
Security Group
(DB-SG)
VPC Endpoint
(S3 Gateway)
10.0.0.0/16
Internet
(For Outbound via
NAT Gateway)
VPC Endpoint
(SQS, SNS, KMS, etc)
(Interface Endpoints)
Storage & Messaging
Amazon S3
(Storage)
Amazon SQS
(Queue)
Amazon MQ
(Message Broker)
Monitoring & Logging
Amazon CloudWatch
(Metrics, Logs, Alarms)
AWS CloudTrail
(API Audit Logs)
Amazon SNS
(Notifications)
Security & Encryption
AWS KMS
(Encryption Keys)
Legend
Traffic Flow
Replication
Logging / Monitoring
Security Groups
Web-SG
Inbound: 80, 443 (from Internet)
Outbound: All
App-SG
Inbound: 8080 (from Web-SG)
Outbound: 3306 (to DB-SG)
Outbound: All (via NAT)
DB-SG
Inbound: 3306 (from App-SG)
Outbound: All
Network ACL Rules (Example)
Public Subnets: Allow 80, 443
Outbound: All
Private Subnets: Deny All Inbound
Allow Required Outbound

1. Architecture Overview

This architecture provides a secure and highly available foundation for a web or API workload on AWS. The design uses managed AWS services where practical, keeps application resources private, and separates edge, application, data and operations responsibilities.

2. AWS Services

Layer	AWS Service	Role
DNS	Amazon Route 53	Domain resolution and health checks
Edge	Amazon CloudFront	Content delivery and caching
Security	AWS WAF	Web request protection
Traffic	Application Load Balancer	HTTPS termination and routing
Compute	Amazon EC2 / ECS	Application workloads
Scaling	Auto Scaling	Elastic capacity and recovery
Database	Amazon RDS	Managed relational database
Storage	Amazon S3	Static content and object storage
Monitoring	Amazon CloudWatch	Metrics, logs and alarms
Audit	AWS CloudTrail	AWS API activity and audit
Secrets	AWS Secrets Manager	Application credentials
Identity	AWS IAM	Access control
Encryption	AWS KMS	Key management and encryption

3. Network Design

VPC Dedicated VPC with controlled routing and clear network boundaries.

Availability Zones Application capacity distributed across at least two AZs.

Private Application Tier Application workloads are not directly exposed to the internet.

Database Tier Database resources remain in the private data tier.

Security Groups Only required ports and source paths are allowed between tiers.

4. Security Design

AWS WAF Filters and protects web traffic before application processing.

IAM Uses roles and least-privilege permissions.

Secrets Manager Keeps sensitive credentials outside application code.

KMS Provides controlled encryption key management.

Security Groups Restrict network traffic between application components.

CloudTrail Provides an audit trail of AWS API activity.

5. Request Flow

User accesses the application domain.

Route 53 resolves the domain.

CloudFront handles cacheable content.

AWS WAF evaluates the incoming request.

Application Load Balancer routes to a healthy private target.

EC2/ECS processes the request.

RDS provides transactional data and S3 provides object/static content.

CloudWatch and CloudTrail provide monitoring and audit visibility.

6. Availability & Scalability

Use multiple Availability Zones for application capacity.

Use Auto Scaling to add or remove application instances based on demand.

Use health checks to remove unhealthy targets.

Use RDS Multi-AZ where database failover is required.

Use CloudFront caching to reduce origin load and improve response times.

7. Monitoring & Operations

Area	AWS Service	Purpose
Metrics	CloudWatch	CPU, latency, errors, throughput and capacity
Logs	CloudWatch Logs	Application and infrastructure troubleshooting
Audit	CloudTrail	API activity and change tracking
Alerts	CloudWatch Alarms	Operational notifications
Backup	RDS / AWS Backup	Recovery and retention

8. Backup & Disaster Recovery

Define RPO and RTO for critical workloads.

Enable automated database backups with appropriate retention.

Test database restore procedures periodically.

Document application and database failover procedures.

Maintain recovery runbooks and validate them through exercises.

9. AWS Cost & Efficiency

Apply tags for application, environment and ownership.

Right-size compute using actual utilization.

Use Auto Scaling to reduce idle capacity.

Use S3 lifecycle policies where appropriate.

Review CloudWatch usage and AWS costs regularly.

10. Production Checklist

- ☐ Multi-AZ deployment validated
- ☐ Private subnet design reviewed
- ☐ Security groups reviewed
- ☐ WAF rules tested
- ☐ HTTPS/TLS enforced
- ☐ IAM permissions reviewed
- ☐ Secrets removed from code
- ☐ Encryption requirements implemented
- ☐ CloudWatch alarms configured
- ☐ CloudTrail enabled
- ☐ Backup and restore tested
- ☐ RPO/RTO documented
- ☐ Cost tagging enabled

11. Architecture Summary

USERS → ROUTE 53 → CLOUDFRONT → WAF → ALB → PRIVATE APP → RDS / S3

AWS architecture designed for availability, security, scalability and operational visibility.

AWS 3-Tier Web Application Architecture

Highly Available | Secure | Scalable | Cost Optimized

1 Route 53 (DNS)
Resolves the application domain name and routes users to CloudFront.

2 CloudFront (CDN)
Caches static content at edge locations to improve performance and reduce latency.

3 AWS WAF
Protects the application from common web exploits, SQLi, XSS, rate limiting, IP blocking and other rules.

4 VPC & Networking
- VPC CIDR: 10.0.0.0/16
- Spans across 2 Availability Zones for high availability.

5 Internet Gateway
Enables communication between resources in the VPC and the internet.

6 Public Subnets
- Hosts Internet-facing resources.
- Contains ALB and NAT Gateway.

7 NAT Gateway
Allows private instances to access the internet for updates and external services securely.

8 Application Load Balancer
Distributes incoming traffic across healthy application servers in multiple AZs.

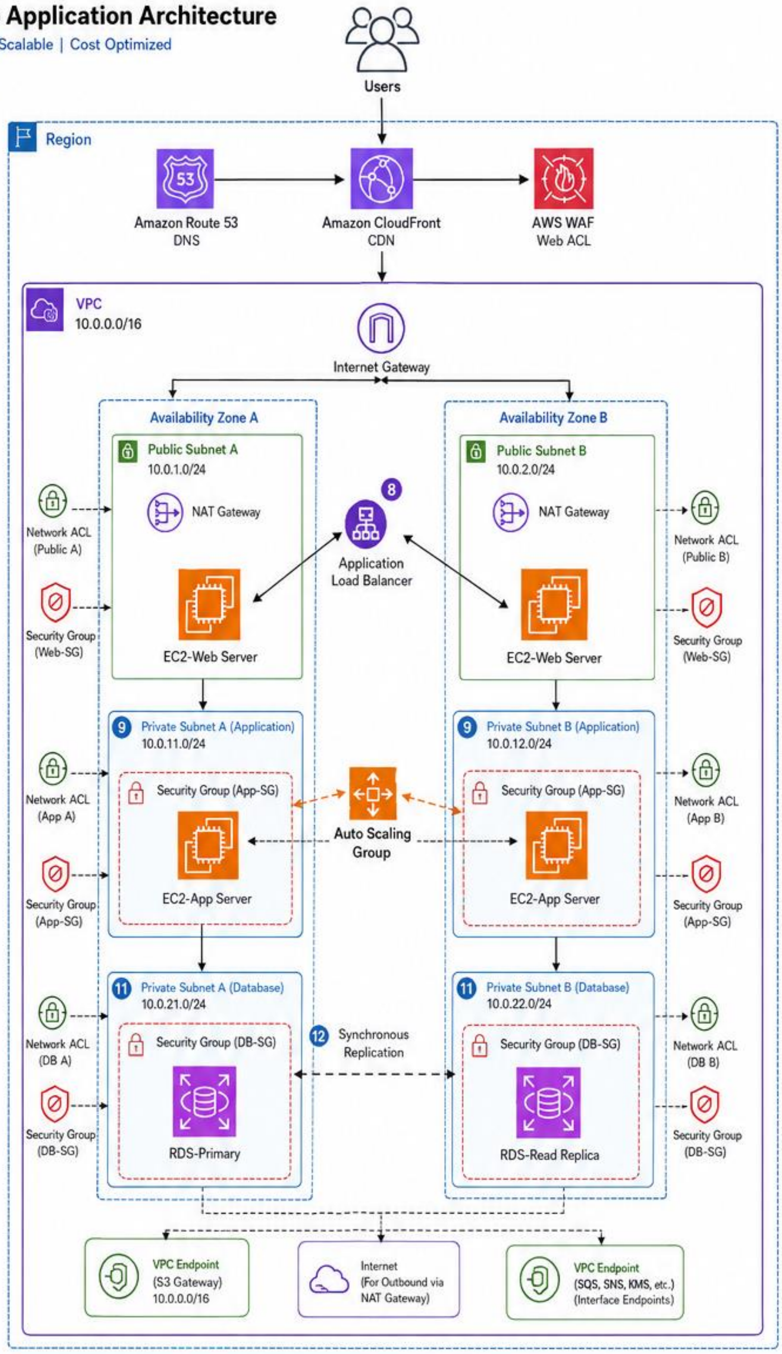
9 Private Application Subnets
- Hosts EC2 application servers.
- Instances are in an Auto Scaling Group for elasticity.

10 Auto Scaling Group
Automatically adjusts the number of EC2 instances based on demand.

11 Private Database Subnets
RDS is placed in private subnets for security. No public access.

12 RDS Multi-AZ
Primary DB in AZ-A and synchronous standby in AZ-B for high availability and automatic failover.

13 VPC Endpoints
Private connectivity to AWS services (S3, SNS, SQS, KMS etc.) without using internet or NAT gateway.



Storage & Messaging

Amazon S3 (Storage)
Stores static files, logs, backups, artifacts, etc.

Amazon SQS (Queue)
Decouples application components.

Amazon MQ (Message Broker)
Managed message broker for reliable messaging.

Monitoring & Logging

Amazon CloudWatch (Metrics, Logs, Alarms)
Collects metrics, logs and creates alarms and dashboards.

AWS CloudTrail (API Audit Logs)
Records all API activities for auditing and compliance.

Amazon SNS (Notifications)
Sends alerts and notifications (Email, SMS, etc.).

Security & Encryption

AWS KMS (Encryption Keys)
Manages encryption keys used by S3, RDS, EBS, SQS, SNS, etc. for data protection.

Legend

- Traffic Flow
- Replication
- Logging / Monitoring
- Network ACL
- Security Group
- EC2 Instance
- RDS Instance
- VPC Endpoint

Security Groups

Web-SG	App-SG	DB-SG
Inbound: 80, 443 (from Internet)	**Inbound:** 8080 (from Web-SG)	**Inbound:** 3306 (from App-SG)
Outbound: All	**Outbound:** 3306 (to DB-SG) & All (via NAT)	**Outbound:** All

Network ACL Rules (Example)

Public Subnets: Allow 80, 443
Outbound: All

Private Subnets: Deny All Inbound
Allow Required Outbound (3306, 443, etc.)

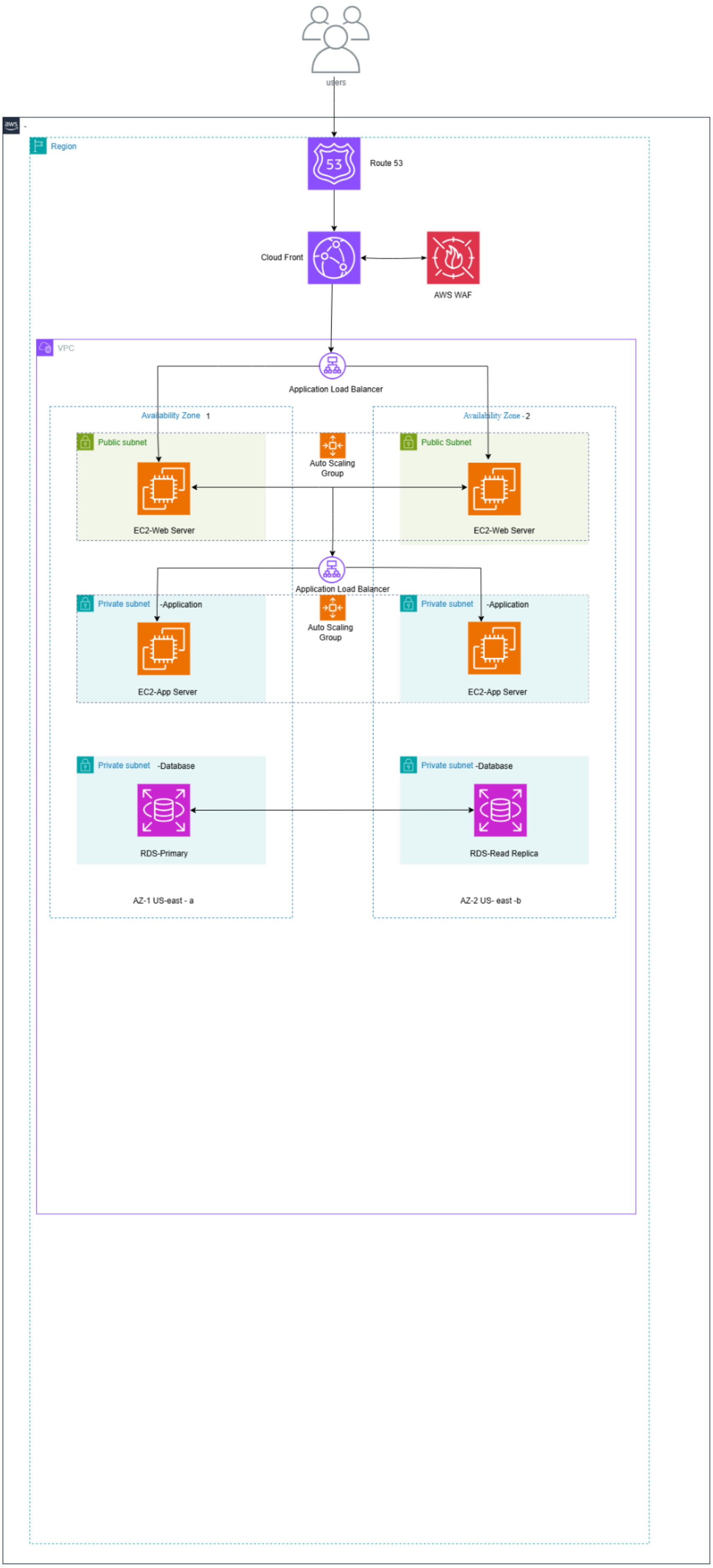
Key Benefits

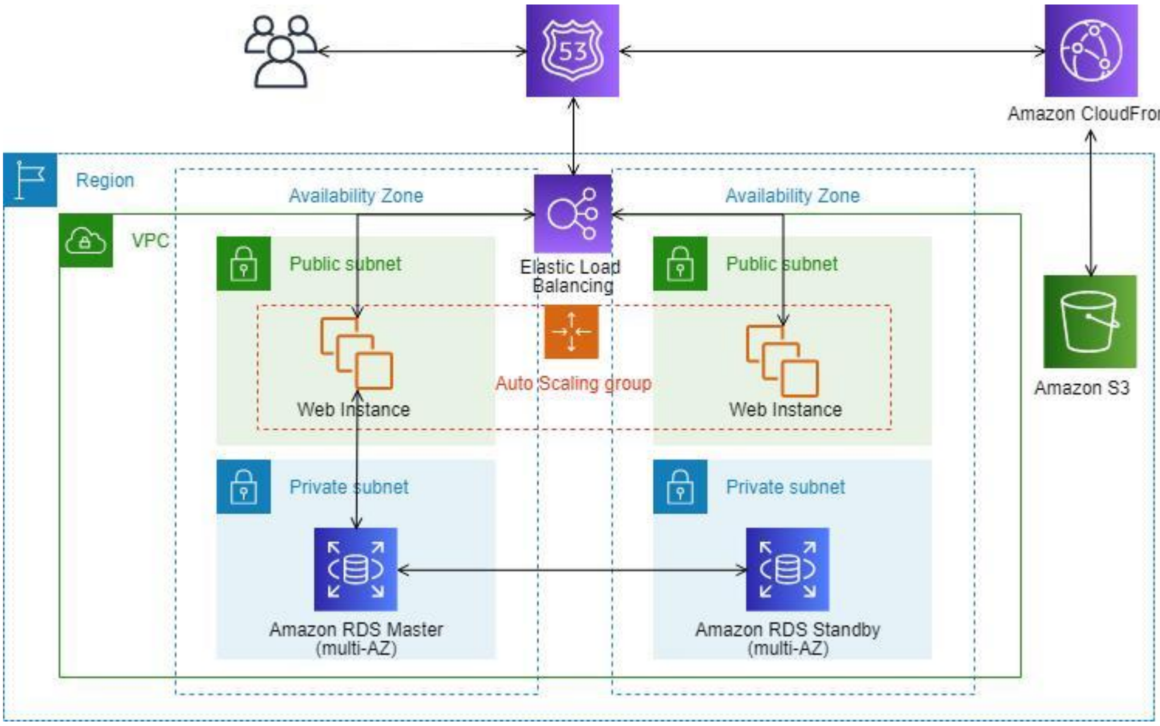
- ✔ **High Availability** (Multi-AZ, RDS Multi-AZ)
- ✔ **Scalability** (Auto Scaling, ALB)
- ✔ **Security** (Private Subnets, SG, NACL, WAF)
- ✔ **Performance** (CloudFront, ALB, Caching)
- ✔ **Reliability** (SQS, MQ, RDS Multi-AZ)
- ✔ **Observability** (CloudWatch, CloudTrail)
- ✔ **Cost Optimization** (S3, VPC Endpoints)

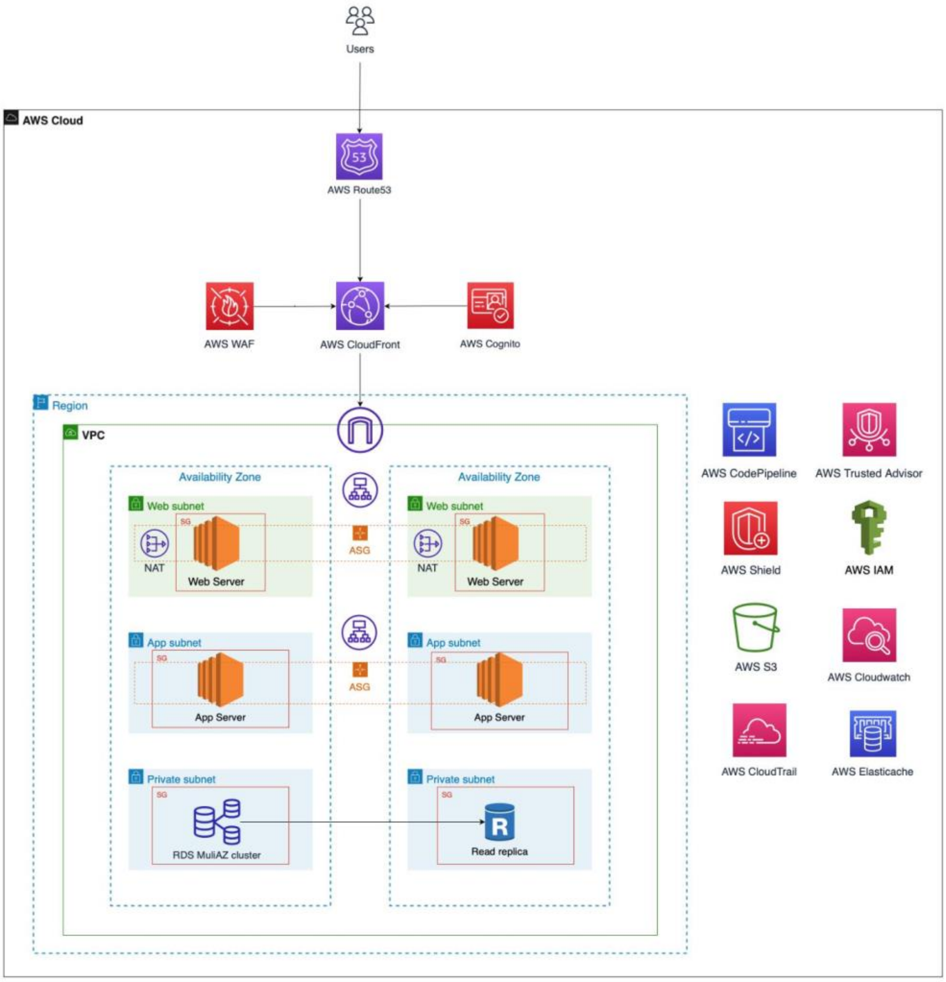
① Users access the application (URL).
② Route 53 resolves DNS and directs to CloudFront.
③ CloudFront delivers cached content or forwards the request to thes ogiгb origin.
④ WAF inspects and allow legitimate traffic.
⑤ Traffic enters the application and ALB, activities and reaches ALB.
⑥ ALB distributes requests to healthy EC2 App servers in Auto Scaling Group across AZs.
⑦ Application servers interact with S3, SQS, MQ, RDS as needed.
⑧ RDS handles data with synchronous replication to standby in other AZ.
⑨ Application logs, metrics, and API activities are sent to CloudWatch and CloudTrail; alerts are sent via SNS.
⑩ All data at rest is encrypted using KMS keys.

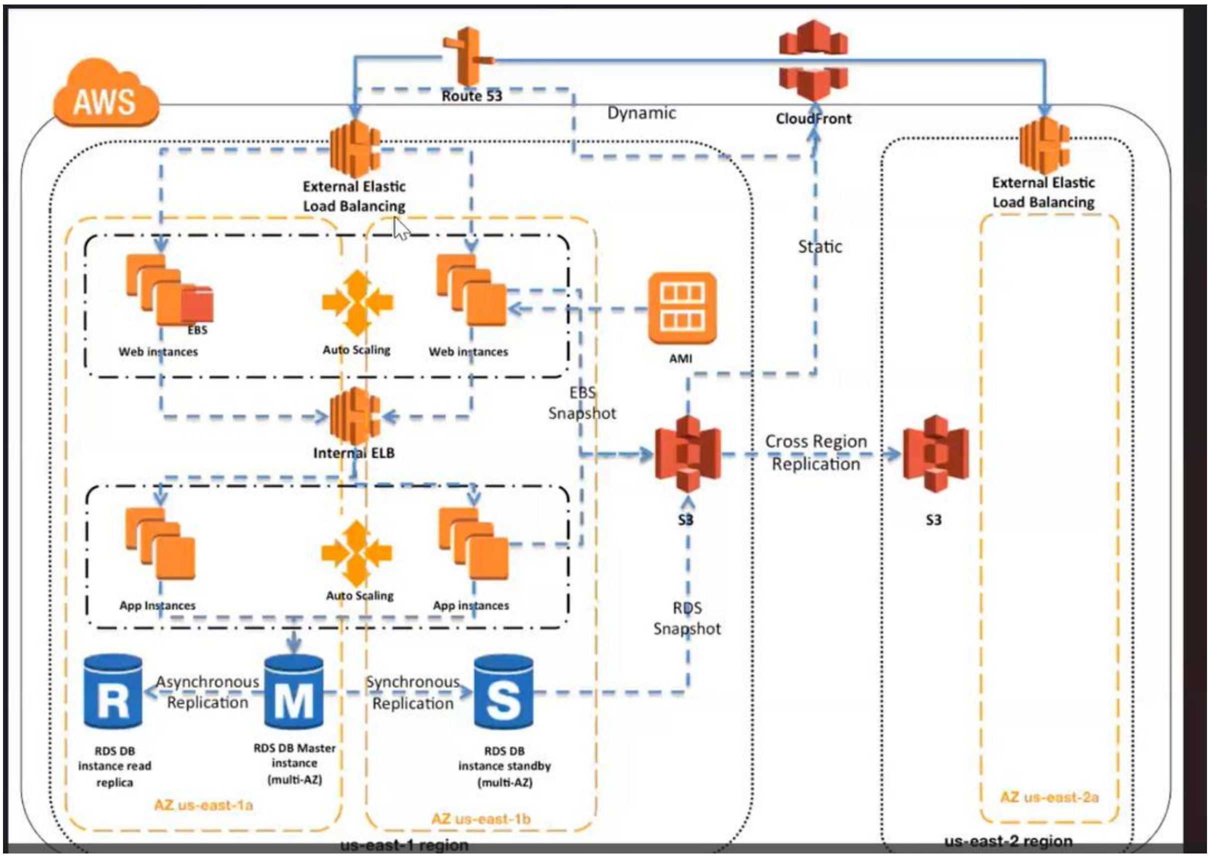
Design Principles Applied

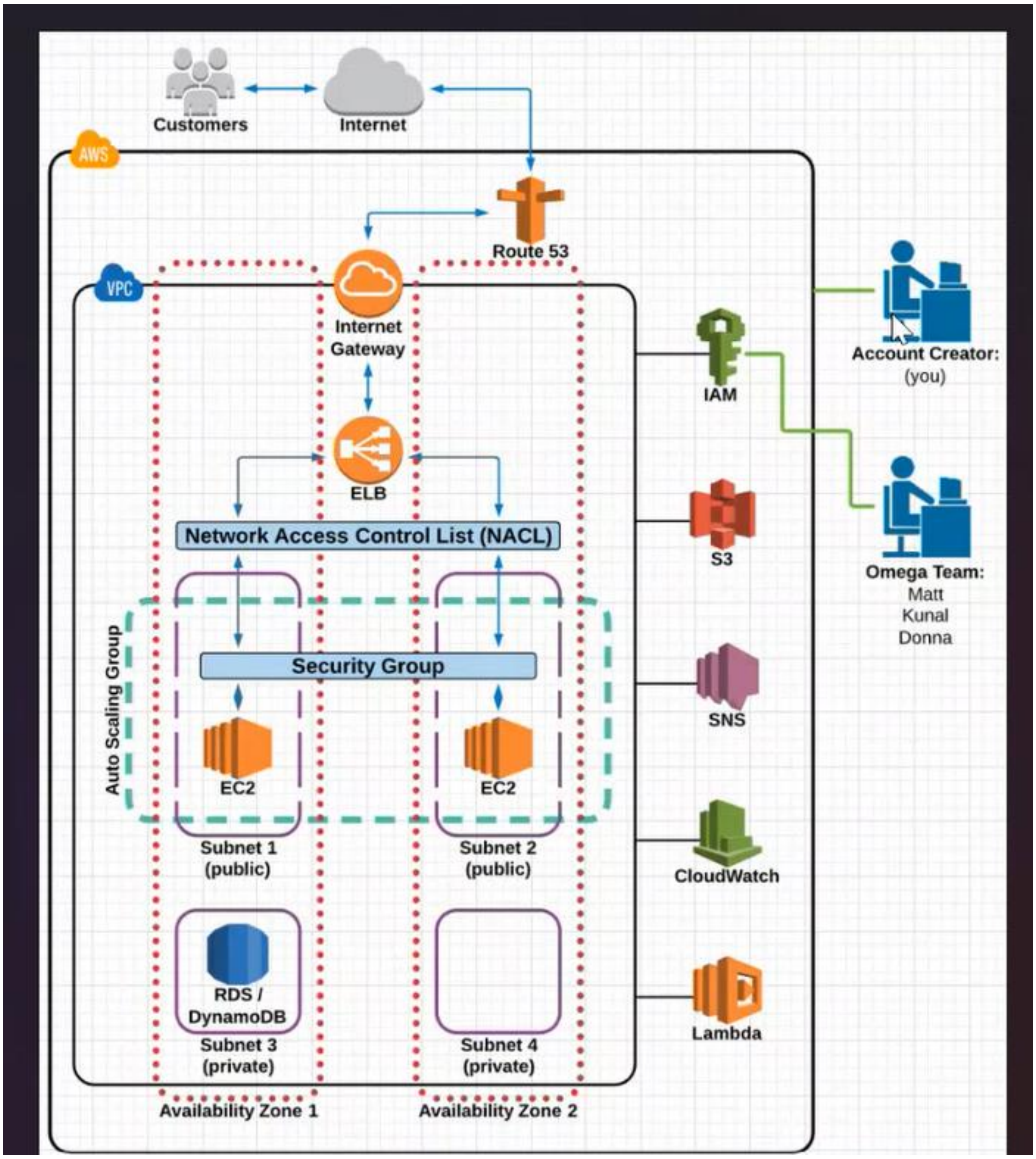
- **Well-Architected Framework:** Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization.
- **Defense in Depth:** Multi-layer security controls.
- **Least Privilege:** Security Groups, IAM, KMS key policies.
- **Fault Tolerance:** Multi-AZ for ALB, and RDS.


users
Region
Route 53
Cloud Front
AWS WAF
VPC
Application Load Balancer
Availability Zone 1
Availability Zone - 2
Public subnet
Auto Scaling Group
Public Subnet
EC2-Web Server
EC2-Web Server
Application Load Balancer
Private subnet -Application
Auto Scaling Group
Private subnet -Application
EC2-App Server
EC2-App Server
Private subnet -Database
Private subnet -Database
RDS-Primary
RDS-Read Replica
AZ-1 US-east - a
AZ-2 US- east -b


Amazon CloudFront
Region
Availability Zone
Availability Zone
VPC
Elastic Load Balancing
Public subnet
Public subnet
Web Instance
Auto Scaling group
Web Instance
Amazon S3
Private subnet
Private subnet
Amazon RDS Master (multi-AZ)
Amazon RDS Standby (multi-AZ)


Users
AWS Cloud
AWS Route53
AWS WAF
AWS CloudFront
AWS Cognito
Region
VPC
Availability Zone
Availability Zone
Web subnet
Web subnet
SG
NAT
Web Server
ASG
NAT
Web Server
App subnet
App subnet
App Server
ASG
App Server
Private subnet
Private subnet
RDS MuliAZ cluster
Read replica
AWS CodePipeline
AWS Trusted Advisor
AWS Shield
AWS IAM
AWS S3
AWS Cloudwatch
AWS CloudTrail
AWS Elasticache


AWS
Route 53
CloudFront
Dynamic
External Elastic Load Balancing
Static
EBS
Web instances
Auto Scaling
Web instances
AMI
EBS Snapshot
Internal ELB
S3
Cross Region Replication
S3
App Instances
Auto Scaling
App instances
RDS Snapshot
Asynchronous Replication
Synchronous Replication
RDS DB instance read replica
RDS DB Master instance (multi-AZ)
RDS DB instance standby (multi-AZ)
AZ us-east-1a
AZ us-east-1b
AZ us-east-2a
us-east-1 region
us-east-2 region



Customers
Internet
AWS
Route 53
VPC
Internet Gateway
ELB
Network Access Control List (NACL)
Security Group
Auto Scaling Group
EC2
EC2
Subnet 1 (public)
Subnet 2 (public)
RDS / DynamoDB
Subnet 3 (private)
Subnet 4 (private)
Availability Zone 1
Availability Zone 2
IAM
S3
SNS
CloudWatch
Lambda
Account Creator: (you)
Omega Team: Matt Kunal Donna